



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
MONDAY TEST

CLASS: X
SUBJECT: COMPUTER APPLICATIONS

TOTAL MARKS: 40
DATE: 3.7.23

Instructions:

The Question Paper comprises of Two Parts. Part-I is based on Multiple Choice Questions and Application questions. Part -II is based on Java Programs. The intended marks for questions are given in brackets []. This paper consists of three printed pages.

PART-I (20 marks)
[Attempt all question]

Question 1

Choose the correct answer from the choices given:

[6×1=6]

i) What is the output of the following:

`System.out.print("India is Great".substring(1,5));`

- | | |
|---------|----------------------|
| a) aidn | b) Compilation error |
| c) ndia | d) No output |

ii) What is the output of the given expression:

`int a=1, b=2, c=3;`

`System.out.print(a++ / - -b - ++c);`

- | | |
|-------|---------|
| a) -3 | b) -3.0 |
| c) 3 | d) 3.0 |

iii) The "if(n%10==0)" statement checks that the number 'n' is _____ with 10.

- | | |
|--------------|--------------|
| a) divisible | b) equal to |
| c) ending | d) beginning |

iv) `System.out.print(Integer.parseInt("2") + Double.parseDouble("3.5"));` will return:

- | | |
|---------|--------|
| a) 23.5 | b) 5.5 |
| c) 5 | d) 5.0 |

v) In which of the following any changes in the formal parameter will not reflect on the actual parameter:

- | | |
|----------------------|-----------|
| a) pass by reference | b) class |
| c) pass by value | d) object |

vi) What will be the output of “Transparent”.compareTo(“Transition”):

- a) 6 b) 8 c) 7 d) -7

Question 2

a) What is the difference between instantiation and initialization? [2]

b) Convert the given snippet to ternary operator: [2]

```
if(name== “Computer Applications”)
s= “You are at the right place”;
else s= “Try again”;
```

c) Write a java statement for each to perform the following: [2]

- i. Find and display the position of the last space in a string
- ii. Find and display the fifth character of the string

d) Predict the output: [2]

```
char ch; int x=100;
do { ch=(char)x;
    System.out.print(ch+" ");
    if(x%10==0)
    break;
    ++x;
}while(x<=100);
```

e) Predict the output: [3]

```
class abc
{ void main()
{ int i,j;
  for(i=1;i<=7;i++)
  { for(j=1;j<i;j++)
    { System.out.print(j);
    } if(i%2==0)
    continue;
    System.out.println();  }}
```

f) Predict the output:

[3]

```
class p1
{ void main()
{ String s1="45", s2="200";
String s3=s2+s1;
int in=Integer.valueOf(s1+s2);
System.out.println(in);
String s4=Integer.toString(in);
System.out.println((s3)+30);
System.out.println("abc"+s4+3);  }}
```

PART-II (20 marks)

[Attempt all questions. Each program should be written using Variable descriptions/Mnemonic Codes so that the logic of the program is clearly depicted]

Question 3

[10]

Design a class to overload a method series() as follows:

1. double series(double n) with one double argument and returns the sum of the series.
sum = $(1/1) + (1/2) + (1/3) + \dots + (1/n)$
2. double series(double a, double n) with two double arguments and returns the sum of the series. sum = $(1/a^2) + (4/a^5) + (7/a^8) + (10/a^{11}) + \dots$ to n terms.
Write a suitable main() method.

Question 4

[10]

Special words are those words which start and end with the same letter.

Example: EXISTENCE, COMIC, WINDOW

Palindrome words are those words which read the same from left to right and vice-versa.

Example: MALYALAM, MADAM, LEVEL, ROTATOR, CIVIC

All palindromes are special words but all special words are not palindromes.

Write a program to accept a word. Check and display whether the word is a palindrome or only a special word or none of them.